Waseem Raza, PhD

Machine Learning Engineer | Healthcare AI & Wearable Intelligence

Healthcare AI | Wearable Sensing | Deep Learning | Reliable ML

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Profile

Machine Learning Engineer and healthcare-AI researcher with a PhD in Electrical and Computer Engineering and experience applying deep learning, time-series modeling, wearable sensing, and reliable ML to real-world health-monitoring problems. Expertise includes Python, PyTorch, TensorFlow, scikit-learn, CNNs, LSTMs, attention, activity recognition, smartwatch IMU analytics, medication-taking detection, personalization, domain adaptation, few-shot learning, cross-user validation, and model evaluation. Complementary industry experience includes production ML, cloud data pipelines, data-quality validation, and multimodal analytics.

Technical Competencies

Healthcare AI & Wearables: Wearable sensing, medication-adherence monitoring, activity recognition, smartwatch IMU data, human-centered sensing, and health-data applications

Deep Learning: CNNs, LSTMs, attention, transfer learning, domain adaptation, representation learning, classification, and sequence modeling

Time-Series & Sensor ML: Windowing, segmentation, feature engineering, temporal modeling, cross-user evaluation, personalization, and continuous-event detection

Reliable & Personalized ML: Distribution shift, few-shot adaptation, fairness-aware evaluation, robustness analysis, LOSO validation, and error analysis

Data & Engineering: Python, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, cloud analytics, data pipelines, APIs, and reproducible experimentation

Research & Evaluation: Experimental design, statistical analysis, sensitivity/specificity, precision/recall/F1, cross-validation, visualization, and technical communication

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Lead end-to-end development and validation of production machine-learning workflows for high-impact operational decision support, including data preparation, model evaluation, error analysis, and reliability testing.
- Design domain-adaptive neural models and reusable preprocessing pipelines to improve model generalization under heterogeneous data distributions.
- Develop multimodal AI workflows that combine structured and unstructured operational data, providing transferable experience for complex real-world data environments.
- Partner with cross-functional stakeholders to translate domain requirements into validated ML workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Developed scalable analytical and ETL pipelines using Python, Azure Databricks, Azure Functions, and Synapse Analytics for large domain datasets.
- Built reusable ingestion, cleansing, transformation, and validation workflows for machine-learning and LLM-assisted analytics.
- Implemented cross-validation, data-quality, and consistency checks to improve reliability and reproducibility of analytical outputs.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Projects & Research

Projects & Applied Research

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*
Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.
Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

Cloud Computer Vision (GCP AutoML / Object Detection) *Guided Project | 2025*
Implemented cloud-based computer-vision workflows for image classification, object detection, and vehicle counting using managed model services and Python inference pipelines. Evaluated preprocessing, inference outputs, class/count aggregation, and model performance across visual-data tasks.
Tools & Methods: Python, Computer Vision, Object Detection, Image Classification, Google Cloud AutoML, Streaming, Preprocessing, Inference, Model Evaluation

OCR & Document Intelligence Extraction Pipeline *Guided Project | 2025*
Implemented a document-intelligence pipeline that converts reports, invoices, and contract-style documents into structured outputs through OCR, layout/table parsing, entity and field extraction, normalization, and validation. Designed the workflow for downstream analytics and data integration rather than isolated text extraction.
Tools & Methods: OCR, Document Intelligence, LLMs, Named Entity Recognition, Table Extraction, Structured JSON, Normalization, Validation, Data Integration

Agentic AI and RAG Applications *Portfolio Project | 2026*
Built grounded GenAI workflows that combine document retrieval, vector search, stateful orchestration, tool-using agents, memory, evaluation, and multi-agent handoffs. Emphasized controllable execution, traceable context flow, and reliable responses for multi-step tasks rather than single unconstrained model calls.
Tools & Methods: Python, LangGraph, LangChain, CrewAI, AutoGen, RAG, Vector Databases, Embeddings, Tool Calling, Agent Memory, Evaluation

Amazon Bedrock Generative AI & Agent Workflows *Certification Project | 2026*
Implemented foundation-model prompting, knowledge-base retrieval, RAG integration, model-customization exercises, workflow automation, and intelligent-agent patterns using Amazon Bedrock. Explored production-oriented cloud GenAI patterns with managed model access, retrieval-backed grounding, and reusable application workflows.
Tools & Methods: Amazon Bedrock, Foundation Models, RAG, Knowledge Bases, LangChain, AI Agents, Prompt Engineering, AWS GenAI Services, Workflow Automation

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*
Designed an AI-ready lakehouse pipeline that prepares structured and semi-structured data for analytics and downstream machine-learning workloads. Implemented repeatable ingestion, cleansing, transformation, validation, and scalable processing patterns suitable for reproducible cloud data workflows.
Tools & Methods: Azure Databricks, Apache Spark, PySpark, ETL/ELT, Data Ingestion, Schema Transformation, Data Validation, Feature Preparation, Cloud Data Engineering

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *2026*
HealthINF | Accepted

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

Certifications & Professional Development

1. Health Data Science Foundation — University of Illinois Urbana-Champaign / Coursera | Completed | March 15, 2026

Focus: Builds foundational skills for working with healthcare data by connecting data-science methods with clinical and health-domain questions. Emphasizes data representation, preprocessing, analysis workflows, interpretation, and practical constraints associated with health datasets.

Tools & Skills: health data science, healthcare datasets, data preprocessing, statistical analysis, data interpretation, reproducible analytics, health-domain reasoning

2. Generative AI for Healthcare — Google Cloud / Coursera | Completed | February 17, 2026 | Certificate

Focus: Explores generative-AI applications in healthcare, including information workflows, domain-aware application design, and opportunities to augment health-data analysis and communication. Emphasizes responsible adoption, usefulness, and evaluation within healthcare constraints.

Tools & Skills: healthcare GenAI use cases, prompt and application design, health-data workflows, responsible AI, domain-specific evaluation, AI adoption considerations

3. Detecting COVID-19 with Chest X-Ray using PyTorch — Coursera | Completed | January 27, 2026

Focus: Applies deep learning to chest X-ray image classification using PyTorch, covering image preprocessing, model training, validation, and interpretation in a healthcare context. Demonstrates an end-to-end computer-vision workflow for medical-image analysis.

Tools & Skills: PyTorch, medical image classification, convolutional neural networks, image preprocessing, training and validation, model evaluation, healthcare AI

4. Introduction to Deep Learning & Neural Networks with Keras — IBM / Coursera | Completed | August 12, 2026

Focus: Introduces deep-learning and neural-network fundamentals using Keras, covering network architecture, training, validation, and model improvement. Connects concepts such as activation functions and optimization with practical neural-network implementation and evaluation.

Tools & Skills: Keras, deep learning, neural networks, model training, validation, activation functions, optimization, model evaluation

5. Machine Learning — Stanford Online / Coursera | Completed | March 29, 2020

Focus: Covers foundational machine-learning methods for supervised and unsupervised problems, including regression, classification, model evaluation, regularization, clustering, and practical model-selection considerations. Emphasizes translating data into predictive models while controlling generalization error.

Tools & Skills: supervised learning, regression, classification, regularization, clustering, model evaluation, bias-variance tradeoffs, machine-learning workflows

6. Introduction to Healthcare Management — Coursera | Completed | March 3, 2026

Focus: Introduces management concepts for healthcare organizations, including organizational structure, stakeholders, service delivery, operations, leadership, and resource coordination. Emphasizes how managerial decisions affect quality, efficiency, and performance in healthcare systems.

Tools & Skills: healthcare management, healthcare operations, organizational structure, stakeholder management, resource coordination, service delivery, performance improvement

7. Python for Data Science, AI & Development — IBM / Coursera | Completed | July 12, 2026 | Certificate

Focus: Builds practical Python skills for data science and AI development, including core programming, data structures, data manipulation, and application-oriented analytical workflows. Emphasizes using Python as a reusable foundation for data processing, modeling, and AI applications.

Tools & Skills: Python, data structures, data manipulation, Pandas concepts, NumPy concepts, data-analysis workflows, AI application development